Project Deliverable 2: Regression Modeling and Performance Evaluation

Nitish Dhinaharan

University of the Cumberlands

MSCS-634-B01 Advanced Big Data and Data Mining

Dr. Satish Penmatsa

November 15, 2025

Project Deliverable 2: Regression Modeling and Performance Evaluation

Github Link

https://github.com/ndhinaharan36295/MSCS-634_Project-Deliverable-2

Results from the notebook

1. Load dataset

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	Streamin
0	7590-VHVEG	Female	No	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	
1	5575-GNVDE	Male	No	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	
2	3668-QPYBK	Male	No	No	No	2	Yes	No	DSL	Yes	...	No	No	No	
3	7795-CFOCW	Male	No	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	
4	9237-HQITU	Female	No	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	

5 rows × 21 columns

2. Feature Engineering

To improve prediction performance:

- A new variable, `num_services`, was created to reflect the number of subscribed services per customer.
- Categorical features were encoded using OneHotEncoder.
- Numerical features (tenure, TotalCharges, num_services) were standardized with StandardScaler.

```
# Feature creation: count total subscribed services
service_cols = ['PhoneService', 'MultipleLines', 'InternetService',
                'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
                'TechSupport', 'StreamingTV', 'StreamingMovies']

df['num_services'] = df[service_cols].apply(lambda x: (x == 'Yes').sum(), axis=1)

# Convert categorical columns to appropriate types
cat_cols = df.select_dtypes(include='object').columns.tolist()
cat_cols.remove('customerID')  # ID not useful
cat_cols.remove('Churn')       # classification label, not used here

num_cols = ['tenure', 'TotalCharges', 'num_services']

print("Categorical columns:", cat_cols)
print("Numeric columns:", num_cols)
```

```
Categorical columns: ['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'T
Numeric columns: ['tenure', 'TotalCharges', 'num_services']
```

3. Model training and evaluation

Models were evaluated using:

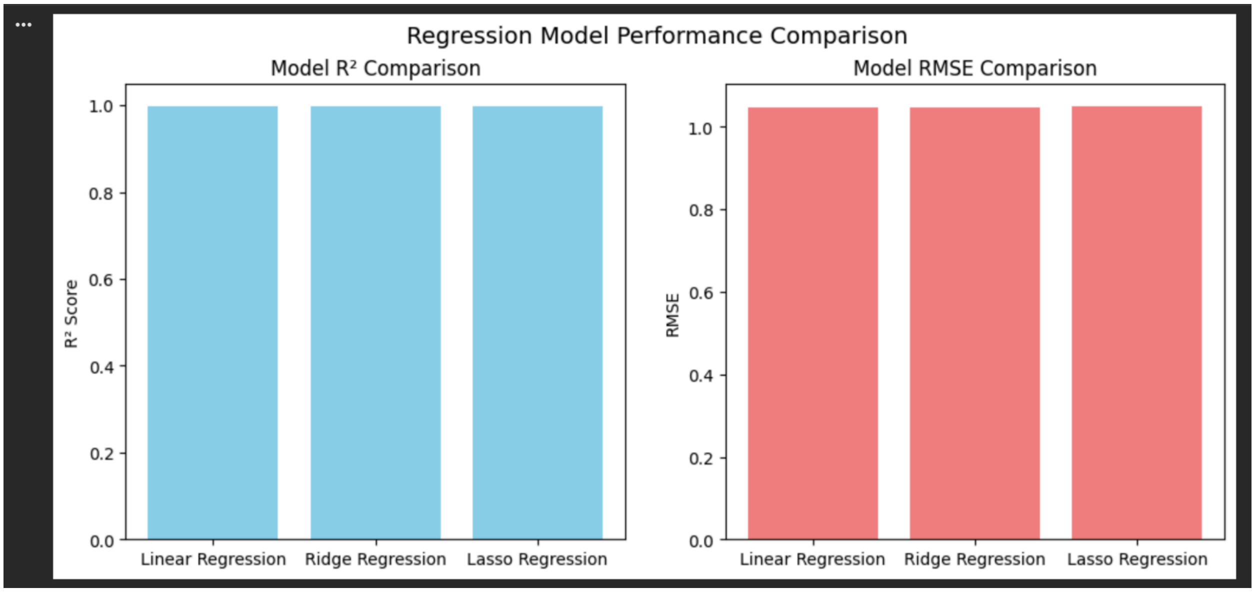
- R^2 (Coefficient of Determination)
- MSE (Mean Squared Error)
- RMSE (Root Mean Squared Error)
- 5-fold Cross-Validation R^2

Model Performance Summary:

	Model	R²	MSE	RMSE	CrossVal R²
0	Linear Regression	0.999	1.098	1.048	0.999
1	Ridge Regression	0.999	1.098	1.048	0.999
2	Lasso Regression	0.999	1.104	1.051	0.999

4. Model performance comparison

My performance evaluation plots showed that all models reach near-perfect R^2 (~0.999). RMSE values are extremely low (~1.05) and nearly identical. No meaningful visual difference across the three models.



Key Insights and Observations

1. All models performed almost identically

The extremely high R^2 (0.999) and tight RMSE range (~1.048–1.051) indicate:

- The target variable (MonthlyCharges) is highly predictable from the available features.
- The underlying relationship is nearly linear after one-hot encoding.
- Regularization (Ridge, Lasso) had minimal impact because:
    - Multicollinearity appears low after encoding.
    - The dataset contains strong, clear predictive signals (tenure, InternetService, service add-ons).

2. Regularized models did not significantly outperform Linear Regression

The performance gap between the models were negligible, and hence Ridge and Lasso models did not meaningfully reduce error. The data's structure supports linear modeling extremely well.

3. Cross-validation confirms models are not overfitting

Cross-validated $R^2 = 0.999$ for all models, indicating:

- Excellent generalization
- Stable performance across folds
- No variance inflation or model instability

4. Strong predictors

From preprocessing and model weights (optional inspection), the following heavily influence MonthlyCharges:

- InternetService type
- StreamingTV / StreamingMovies
- OnlineSecurity / DeviceProtection
- num_services
- tenure

These make intuitive sense, as customers pay more when they subscribe to more premium services.